

GeoDraw User Guide

End-User Manual

Version: Stable Desktop Build

March 9, 2026

Contents

1 Purpose

GeoDraw is an interactive geometry app for constructing points, lines, circles, angles, sectors, and polygons, then exporting the figure to TikZ (`tkz-euclide` style workflow).

2 Interface Overview

The window is divided into:

- **Left Toolbar:** tool groups (Move, Points, Lines, Angle, Circles, Styles).
- **Center Canvas:** drawing and interaction area.
- **Top Actions:** file controls (Open, Save, Save As), Undo/Redo, Fit View.
- **Right Sidebar:** **Objects** tab and **Export** tab.
- **Bottom Command Bar:** expression evaluation and command-driven construction.

3 File Operations

3.1 Open / Save / Save As

- **Open:** load `.geodraw` or `.json` snapshot.
- **Save:** overwrite current file if already opened/saved once.
- **Save As:** choose new name/location.
- You can also drag and drop a `.geodraw/.json` file onto the canvas to open it.

3.2 Supported file types

- `.geodraw`
- `.json` (same snapshot structure)

4 Navigation and Global Controls

4.1 Mouse

- Drag points to move them.
- Pan the canvas in Move mode.
- Use mouse wheel/trackpad to zoom.

4.2 Keyboard shortcuts

- **Ctrl/Cmd + Z**: Undo
- **Ctrl/Cmd + Shift + Z** or **Ctrl/Cmd + Y**: Redo
- **Shift + F**: Fit View
- **Delete/Backspace**: delete selected object
- **Shift + ArrowUp/Shift + ArrowDown** (Object Browser): extend linked selection to previous/next object of the same type
- **Esc**: cancel Copy Style mode / clear pending selections

4.3 Fit View

Use **Fit View**  from the top action row (next to Undo/Redo), or press **Shift + F**.

- Re-centers the camera to include current scene geometry.
- Applies padding so objects are not touching viewport edges.
- If no geometry is available, camera stays unchanged.

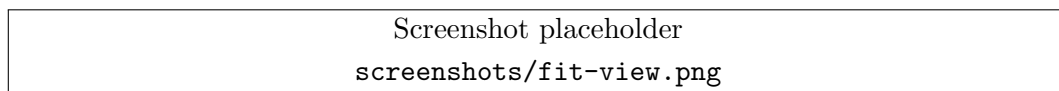


Figure 1: Fit View action and resulting re-centered camera.

5 Tool Palette

Long-press or right-click a group button to open its flyout tools.

5.1 Move Group

-  **Move / Select** (V): select and manipulate objects.
-  **Export Clip Rectangle**: create rectangular clip for export.
-  **Export Clip Polygon**: create polygon clip for export (close by clicking near the first vertex).

5.2 Points Group

-  **Point** (P): create/select points.
-  **Midpoint** (M): midpoint from two points or from a segment.

5.3 Transform Group

-  **Translate**: create translated copy from vector AB.
-  **Rotate**: rotate object around selected center.
-  **Reflect**: reflect object across line/segment/point.
-  **Dilate**: scale object from selected center.
- **Invert**: invert a source line/circle with respect to a selected inversion circle.

All transform tools preserve source styling on the transformed image (stroke/fill/marks/arrows/labels).

5.4 Lines Group

-  **Segment (S)**
-  **Line Through 2 Points (L)**
-  **Perpendicular Line**
-  **Parallel Line**
-  **Tangent Line**
-  **Internal Angle Bisector**

5.5 Angle Group

-  **Angle**: select 3 points (A, vertex B, C).
-  **Angle with Fixed Value**: define expression and direction (CW/CCW), then construct.

5.6 Circles Group

-  **Circle Center + Point (0)**
-  **Circle through 3 Points**
-  **Circle with Fixed Radius**
-  **Circular Sector**
-  **Polygon**
-  **Regular Polygon**

5.7 Styles Group

-  **Copy Style (C)**: pick a source style, then apply to targets.
- **T Label Tool**: place text labels on the canvas.

6 Objects Tab (Right Sidebar)

6.1 Top toggles

- **Grid**: show/hide grid.
- **Axes**: show/hide axes.
- **Snap**: snap free-point creation to grid intersections.
- **Dependency Glow**: highlight dependency categories visually.

6.2 Object Browser

- Tabs: All, Points, Lines, Shapes, Angles, Labels, Numbers.
- Auto-follow: selected object switches to its tab automatically.
- Pin button (lock icon): freeze current tab and disable auto-follow until unpinned.
- Click row to select object.
- Link button (chain icon): link/unlink object for batch style/delete operations.
- Linking is type-restricted: linked objects must be the same type (point with point, line with line, etc.).
- **Shift + ArrowUp/Shift + ArrowDown**: add previous/next same-type row to linked selection.
- Use the row checkbox (right side) to toggle object visibility.

7 Properties Panel

When an object is selected, the panel shows editable properties.

7.1 Linked object editing

- Link multiple same-type objects from Object Browser.
- For linked objects, style/field edits in the Properties panel apply to all linked objects of that type.
- The delete action becomes **Delete All** and removes the linked selection.

7.2 Point properties

- Rename point.
- Position readout.
- Visual style: shape, size, stroke/fill, label style.

7.3 Line/Segment properties

- Stroke color, width, dash, opacity.
- Segment marks (|, ||, |||, etc.) and arrow marks (end/mid).
- **Convert to Segment** (line only): hide source `Line(A,B)` and create `Segment(A,B)` with matching style/label fields.
- If multiple lines are linked, conversion applies to all linked convertible lines.

7.4 Arrow controls

Arrow styling  is available in Segment, Circle, and Angle (arc) style panels.

- Multi-arrow list editor: add, duplicate, remove, and select arrow entries.
- Direction options: forward/backward and paired directions.
- Tip style: Stealth, Latex, Triangle.
- Placement and distribution controls (single/multi, plus segment end/mid modes where applicable).
- Appearance controls: color, width, size, and arrow length.

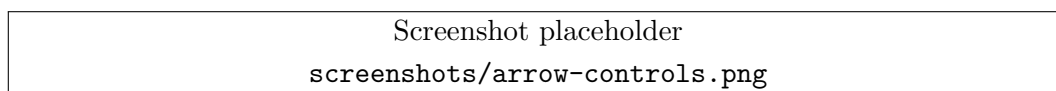


Figure 2: Arrow list editor in object style properties.

7.5 Circle/Polygon/Sector properties

- Stroke and fill settings.
- Fill opacity.
- Fill pattern and pattern color.

7.6 Angle properties

- Arc radius, stroke, fill.
- Label/value visibility and placement.
- Mark styles:
 - Non-right: vanilla arc, bar marks (|, ||, |||), double/triple arc, none.
 - Right-angle status aware options for exact/proven right angles.

7.7 Marking controls

Marking tools  support both segment congruence marks and angle arc marks.

- **Segment marks:** symbol, distribution, position, size, width, and color.
- **Angle marks:** single/double/triple arc count, bar symbol, position, size, and color.
- Both use list-based editors, so multiple mark entries can be layered.

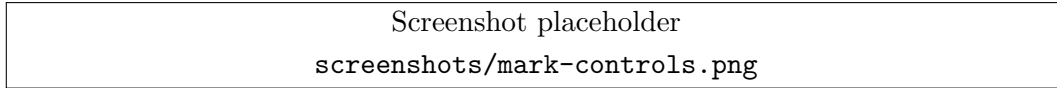


Figure 3: Segment and angle marking controls in the properties panel.

7.8 Tool info panels

Special tools (Fixed Angle, Fixed Radius Circle, Export Clip, Copy Style) show context-specific instructions and inputs.

8 Command Bar

Command Bar supports expression evaluation, object creation commands, and assignments.

8.1 Input behavior

- **Enter:** run command
- **Esc:** clear input
- **Up/Down:** command history

8.2 Expression examples

- $5*5$
- $\sin(\pi/6)$, $\text{atan2d}(4,3)$
- $\text{Distance}(A,B)$

8.3 Supported construction commands

- **Point/transform:** $\text{Point}(x,y)$, $\text{Midpoint}(A,B)$, $\text{Midpoint}(s)$, $\text{Incenter}(A,B,C)$, $\text{Orthocenter}(A,B,C)$ (Ortho), $\text{Centroid}(A,B,C)$, $\text{Circumcenter}(A,B,C)$, $\text{Translate}(P,A,B)$, $\text{Rotate}(P,0,\text{expr}[,CW|CCW])$, $\text{Dilate}(P,0,k)$, $\text{Reflect}(P,l|s|O)$.
- **Line/segment:** $\text{Line}(A,B)$, $\text{Line}(x1,y1,x2,y2)$, $\text{Perpendicular}(P,l)$, $\text{Parallel}(P,l)$, $\text{Tangent}(P,c)$, $\text{AngleBisector}(A,B,C)$, $\text{Segment}(A,B)$.
- **Angle/polygon:** $\text{Angle}(A,B,C)$, $\text{AngleFixed}(V,A,\text{expr}[,CW|CCW])$, $\text{Sector}(O,A,B)$, $\text{Polygon}(A,B,C, \dots)$, $\text{RegularPolygon}(A,B,n[,CW|CCW])$.
- **Circle:** $\text{Circle}(x,y,r)$, $\text{Circle}(O,A)$, $\text{Circle}(O,r)$, $\text{Circle3P}(A,B,C)$ (alias: $\text{CircleThreePoint}(A,B,C)$).
- **Numeric query:** $\text{Distance}(\dots)$, $\text{Area}(\dots)$, $\text{Perimeter}(\dots)$, $\text{Inradius}(A,B,C)$, $\text{Circumradius}(A,B,C)$ return scalar values (no object created).

8.4 Assignments

- **Scalar:** $n_1 = 2.023$
- **Named object:** $P = \text{Point}(1,2)$, $l = \text{Line}(A,B)$, $c = \text{Circle}(O,r_1)$
- **Point-expression assignment (vector style):** $X = A + B$

8.5 Important assignment rule

Assignment is selective and fail-closed:

- Existing constant numbers are updated in-place by scalar assignment.
- Existing free points are updated in-place by point assignment.
- Existing aliases can be redefined only with compatible constructor types.
- Non-free points, non-constant numbers, and incompatible alias redefines are rejected.

9 Export Tab

9.1 TikZ export options

- **Use Current View:** export based on current camera viewport.
- **Use Export Clip selection:** apply clip polygon from Export Clip tool.
- **Match canvas size conversion**
- **Label glow**
- Scale controls:
 - Global Scale
 - Point Scale
 - Line Scale
 - Label Scale

9.2 Preview window (PDF + editable TikZ)

Press **Preview**  in Export to open the dedicated TikZ preview window.

- Left pane: compiled PDF preview from the current TikZ source.
- Right pane: editable `tikzpicture` code.
- **Optional Preamble** section for custom preamble lines (for example `\pagecolor{...}`).
- Find/replace tools with case-match option.
- **Update PDF** recompiles, and **Copy Edited TikZ** copies the edited code.
- Right-click inside PDF preview to save snapshot as **PDF**, **SVG**, or **PNG**.
- Compiler log panel is available for diagnostics when compilation fails.

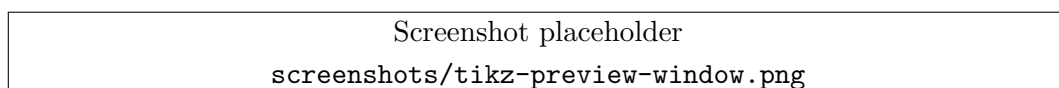


Figure 4: TikZ Preview window with PDF pane, code editor, and compiler log.

9.3 JSON export

- Export model snapshot as JSON.
- Optional debug world coordinates.

10 Typical Workflows

10.1 Workflow A: classic construction

1. Create base points with Point tool.
2. Build segments/lines/circles.
3. Add angles/sectors/polygons.
4. Tune style from Properties panel.
5. Export TikZ.

10.2 Workflow B: command-driven

1. Create anchor points from Command Bar.
2. Create lines/circles/segments via named references.
3. Assign constants and reuse them in commands.
4. Export.

11 Troubleshooting

11.1 I cannot select a tool variant

Long-press or right-click the group button to open its flyout.

11.2 Clip export does not apply

Ensure you closed the clip polygon (click near first point), then enable **Use Export Clip selection** in Export tab.

11.3 Command reports unknown/ambiguous point

- Verify point labels exist.
- Avoid duplicate labels.

11.4 Object is hidden unexpectedly

Check per-object visibility checkbox in Object Browser.

12 Current Limits (User-visible)

- Tangent assignment is intentionally blocked because `Tangent(P,c)` may create multiple lines.
- Alias-based commands (`Midpoint(s)`, `Perpendicular`, `Parallel`, `Tangent`) require prior named aliases.
- Some advanced symbolic redefine workflows are intentionally fail-closed.

A Screenshot Appendix Template

Use this template to insert screenshots consistently.

A.1 Main Layout

A.2 Tool Flyout

A.3 Object Browser

A.4 Properties Panel

A.5 Command Bar

A.6 Export Panel + Clip

A.7 TikZ Output